

CS303E Mock Exam 2
Dr. Bill Young § Spring 2025

Name: _____ EID: _____

Read the questions carefully, and answer each question in the space provided. If you like, you can use scratch paper to do your work, but copy your answers neatly and legibly onto the test paper. Only answers recorded on the test paper will be graded. Don't write in the spaces marked "Page Total" at the bottom of each page. Note: points as listed on the exam sum to 100, but your mock exam grade will be scaled to a 10-point scale.

1. (10 points: 1 point each) The following are true/false questions. **Write either T or F in the boxes at the bottom of page 1.** If there's any counterexample, it's false.
 - (a) If the `__str__` method isn't explicitly defined for a class, python will raise an error if you try to print an object of that class.
 - (b) Within a class, `self.__score` is a private variable.
 - (c) If we execute the assignment `s = "call"`, then afterward we can change the string to `"tall"` by adding the line `s[0] = 't'`.
 - (d) For any given string `s`, the function calls `s.split()` and `s.split(" ")` will return the same list.
 - (e) The expression `bool([])` is equivalent to `False`.
 - (f) If `L1` and `L2` are two lists with the same elements, but different orders, then `L1 == L2` evaluates to `True`.
 - (g) If you pass a mutable item as a parameter, it can be changed by the function.
 - (h) You cannot have both keyword and non-keyword arguments in a function call.
 - (i) You have to declare a variable in Python before you can use it.
 - (j) The loop condition in a `while` loop is evaluated before each iteration.

[illegible]

2. (10 points: 1 point each) The following are short answer questions. Fill in the word or phrase that *best* matches the description provided. If there's a small line in the question, the answer is the word you might write on that line to complete the sentence. *But write it on the long line.*
- (a) _____ Unlike strings which cannot be changed after they're constructed, list objects are _____.
 - (b) _____ This value is returned implicitly from functions with no explicit `return` statement.
 - (c) _____ In the function call `f(2, 5, c=4, d=4)`, while the values for `c` and `d` are passed as keyword arguments, the values 2 and 5 are this type of argument.
 - (d) _____ Object attributes beginning with two underscores are _____ attributes.
 - (e) _____ This function is called whenever a new object is created.
 - (f) _____ Strings are sequences of _____.
 - (g) _____ If `X` evaluates to `True`, you wouldn't evaluate `Y` in `(X or Y)` because of _____ evaluation.
 - (h) _____ The Unicode character encoding was defined such that _____ (the character encoding we've mainly used in this class) is a subset.
 - (i) _____ A contiguous subsequence (e.g., `L[:2]` from list `L` or `S[3:9]` from string `S`) is called a _____.
 - (j) _____ It's easy to differentiate special, so-called "magic methods", which aren't intended to be called directly, within a class, in part because their names begin and end with two _____.

Questions 3–12 are multiple choice. Each counts 1 point. **Write the letter of the BEST answer in the box on the next page. Please write your answer in UPPERCASE. Each problem has a single answer.**

3. Which operator can be used to compare two values?
- A. = B. <> C. == D. >< E. all are legal
4. Which statement is used to stop a loop?
- A. continue B. break C. return D. stop E. exit
5. The default return value of a function is:
- A. 0 B. 1 C. False (D) None E. "None"
6. Which of the following statements accurately describes the mutability of the `int`, `str`, and `list` data types?
- A. All three are mutable
B. All three are immutable.
C. Mutable: `int`, `str`. Immutable: `list`
D. Mutable: `list`. Immutable: `int`, `str`
7. What value of X would make the following True: (True and False or X)
- A. True B. False (C) both work (D) neither works
8. It's not a good idea to use `==` to compare two floats because:
- A. there's nothing wrong with it
B. `==` only works for integers
C. floats are often stored as approximations
D. `=` should be used to compare values
E. there are too many floats to think two might be equal
9. Which type of error, if any, is present in the following code to sum adjacent products in a list of integers? For example, `sumAdjacentProducts([1, 6, 3, 8])` should return $1 \cdot 6 + 6 \cdot 3 + 3 \cdot 8 = 48$.
- ```
def sumAdjacentProducts(L):
 tot = 0
 for i in range(len(L)):
 tot += L[i - 1] * L[i]
 return tot
```
- A. Syntax Error        B. Runtime Error    C. Logic Error        D. No Error



Questions 13–17 require you to trace the behavior of some Python code and identify the output of that code. For each question, write the output for the code segment in the provided box. If it doesn't print anything or gives an error, say so.

```
13. s = "what a beautiful day"
 a_count = 0
 for ch in range(len(s)):
 if ch == "a":
 a_count += 1
 print(a_count)
```

```
14. words = ["banana", "BANANA", "ban", "bandana", "Alabama"]
 words.sort()
 print(words)
```

```
15. def f(lst, j):
 val = lst[j]
 i = j + 1
 while (i < len(lst)) and (lst[i] < val):
 lst[i - 1] = lst[i]
 i += 1
 lst[i - 1] = val
 return lst
```

```
print(f([7, 2, 14, 6, 9, 13, 15, 19], 2))
```

16. *This function is recursive, so feel free to skip this one.*

```
def f(s):
 if s == "":
 return {}
 res = f(s[1:])
 if s[0] not in res:
 res[s[0]] = 0
 res[s[0]] += 1
 return res

print(f("an assassin sins"))
```

```
17. class Pokemon:
 def __init__(self, number, name):
 self.number = number
 self.name = name

 def __str__(self):
 return "This is " + self.name + " with Pokedex id " \
 + str(self.number)

p1 = Pokemon(7, "Squirtle")
p2 = Pokemon(7, "Squirtle")

print(p1 == p2, end=" ")
print(p1.number == p2.number, end=" ")
print(str(p1) == str(p2), end=" ")
```

18. (10 points: 1 point each) The following questions require you to write a *Python expression* that returns the indicated value. You can assume that any modules you need have been imported. Note that this asks for a single, one-line expression for each question, not a longer program fragment. So, you should not have any assignments, loops, or if statements. Hint: if you think you need a loop, try a list comprehension instead.

- (a) \_\_\_\_\_  
a list of the squares of all integers 1 through 100 (i.e. [1, 4, 9, ..., 9801, 10000])  
(Hint: use a conditional expression.)
- (b) \_\_\_\_\_  
for a given string `s`, a list containing all lowercase vowels in `s` (vowels are 'a', 'e', 'i', 'o', and 'u')
- (c) \_\_\_\_\_  
the first half of the even-length, non-empty string `s`
- (d) \_\_\_\_\_  
for a given non-empty list of integers `L`, the number of even integers in `L`
- (e) \_\_\_\_\_  
for a given non-empty list of floats `L`, the average of the values in `L`
- (f) \_\_\_\_\_  
for a given string `s`, `s` repeated 100 times
- (g) \_\_\_\_\_  
for a given non-integer float `num`, the integer immediately above `num`; e.g. if `num` is 3.54, the expression would evaluate to 4
- (h) \_\_\_\_\_  
the value of pi to five decimal places, as a string, displayed right-justified in a field of width 10
- (i) \_\_\_\_\_  
for a given positive float `num`, the part of `num` that comes after the decimal point (e.g. if `num` is 3.14, the expression would give 0.14).
- (j) \_\_\_\_\_  
the distance between points (a, b) and (c, d) (Hint: the distance is the square root of ( (a minus c) squared plus (b minus d) squared))

19. The following 10 questions (worth 1 point each) require you to evaluate a Python expression in the left hand column. For each question, write what the expression evaluates to on the provided line. If evaluation results in an error, write “error”; you don’t have to identity the specific type of error.

**You must show a value of the appropriate type. For example, 7.0 rather than 7 for a float and "7" instead of 7 for a string. Answers that do not indicate the data type correctly are wrong.**

- |                                   |       |
|-----------------------------------|-------|
| (a) "texas longhorns"[-10::2]     | _____ |
| (b) "a!!b!cde".split("!")         | _____ |
| (c) " \nxyz abc ".lstrip()        | _____ |
| (d) "aababcbabcd".rindex("b")     | _____ |
| (e) "wow" >= "wow wow"            | _____ |
| (f) [2, 5, 1].append(7)           | _____ |
| (g) [(x*x - 1) for x in range(5)] | _____ |
| (h) min( "abcDEFghi" )            | _____ |
| (i) ord(chr(15))                  | _____ |

20. (10 points) Fill in the code (on the next page) for the **Square** class with private data members `x` and `y` coordinates of the upper left corner, and the side length. See the sample behavior below.

|                                  |                                   |
|----------------------------------|-----------------------------------|
| >>> from Square import *         | >>> sq2.getArea()                 |
| >>> sq1 = Square(10)             | 25                                |
| >>> print(sq1)                   | >>> sq2.moveTo(7, -2)             |
| Square at (10, 0) with side of 1 | >>> sq2.resize(10)                |
| >>> sq2 = Square(2, 3, 5)        | Resizing square; new side = 10    |
| >>> print(sq2)                   | >>> print(sq2)                    |
| Square at (2, 3) with side of 5  | Square at (7, -2) with side of 10 |
| >>> sq2.getCoordinates()         | >>> sq2.getArea()                 |
| (2, 3)                           | 100                               |

```
class Square:
 def __init__(self, x = 0, y = 0, side = 1):
 # create a new square with coordinates and side (with defaults)

 def __str__(self):

 def getCoordinates(self): # return coordinates as a tuple

 def getArea(self): # return the area

 def moveTo(self, x, y): # change coordinates

 def resize(self, side): # change side length, print message
```

21. (10 points) Fill in the code for function `merge` that takes two sorted (in ascending order) lists `L1` and `L2` of integers (not necessarily the same length). Your function should return a new sorted list of integers containing all of the integers from the original two lists. Duplicates are OK. You must do this by merging the input lists, not by re-sorting the list. Here is some sample behavior:

```
>>> merge([], [1, 2, 4, 6, 8, 9])
[1, 2, 4, 6, 8, 9]
>>> merge([1, 2, 4, 6, 8, 9], [])
[1, 2, 4, 6, 8, 9]
>>> merge([1, 1, 2, 5, 8, 10], [3, 5, 6, 7, 8, 12, 13, 19, 28])
[1, 1, 2, 3, 5, 5, 6, 7, 8, 8, 10, 12, 13, 19, 28]
```

*This is a bit trickier than what would be on the actual exam. Create a new list `newList` by adding on elements from either `L1` or `L2` as appropriate. Define variables `i1` and `i2` into `L1` and `L2` respectively, both initially 0. If either reaches the end of that list, then extend `newList` with the remaining elements from the other list. Think about what slice of the other list is needed for that.*

```
def merge(L1, L2):
```